USER REQUIREMENT SPECIFICATIONS

ICOMS 2.0

Module 1 : System Setup

Contents

1.0 DOCUMENT PURPOSE

The ICOMS 2.0 system is built to provide a single, auditable system for outage request intake and lifecycle management. It follows the scope of outage request, study/history, daily/weekly dockets, approvals and change requests, execution, extensions, and closing outages (planned, unplanned, forced, emergency) across regions and stations, with role and region-based controls, SLD and Conflicting Lines detection and support, Commissioning Memo generation, KIV handling, Handover report, and statistics/analytics.

The purpose of this document is to define the functional and non-functional requirements of the ICOMS 2.0 system, a reinvention of the ICOMS legacy system. It captures the needs of three primary user classes — **Requestor/Planner, TOMS, and GNC** — and provides a unified framework for outage request creation, study, approval, Authorisation, execution, and closure. The document ensures clarity of roles, workflows, and system behaviour to support reliable outage management across regions, stations, and equipment.

A redevelopment of the legacy ICOMS system, ICOMS 2.0 is designed for outage management across the TNB network. It is built for users to request outages for various jobs which will be reviewed, studied and eventually Approved or Rejected. ICOMS 2.0 will take the functionality of the original ICOMS system and add the required enhancements to allow for additional work scope and better user experience.

2.0 OBJECTIVES

The objective of this document is to define the user requirements for the system setup and configuration of the ICOMS 2.0 system. This includes the administrative functions required to initialize and maintain the core system data, user roles, permissions, and operational configuration required for the system to function.

This document specifies the configuration capabilities available to administrative users such as SysAdmin, GNM ADMIN, and GNC Admin to ensure that the system is properly structured before operational use.

The objectives of the system setup are as follows:

1. To establish a structured role-based and zone-based access control framework for all users of the system.
2. To define the processes for user account creation, role assignment, and permission management within the system.
3. To configure and maintain the organizational and geographical structure of the network including zones, locations, organizations, and stations.
4. To set up and maintain voltage levels, equipment types, and equipment records used within the outage management system.
5. To configure system data required for outage operations including off-points, dropdown values, project listings, and authorization personnel.

6. To define the configuration of outage type rules based on work type, duration, and voltage level.
7. To enable administrative setup of transmission lines and conflicting line relationships used for outage validation.
8. To ensure that all required system master data and configuration parameters are established and maintained by authorized administrative users.

3.0 SCOPE

This document covers the **system setup and administrative configuration requirements** for the ICOMS 2.0 system. The scope includes the configuration of system master data, administrative controls, and operational parameters required for the system to support outage management activities.

The scope of this document includes the following areas:

1. **User and Role Management**
   Setup and management of user accounts, roles, permissions, and zone-based access control.
2. **Zone, Organization, and Station Configuration**
   Configuration of geographical zones, locations, organizations, and stations used within the system.
3. **Voltage, Equipment Type, and Equipment Setup**
   Configuration of voltage levels, equipment types, and the creation and management of equipment records.
4. **Off-Point Management**
   Setup and management of permanent and temporary equipment off-points.
5. **System Configuration Data**
   Management of configurable system dropdown values and operational parameters.
6. **Project Management Setup**
   Creation and maintenance of project records associated with outages.
7. **Outage Type Configuration**
   Configuration of system rules used to automatically determine outage types based on defined criteria.
8. **Authorisation Personnel Setup**
   Maintenance of authorised personnel used for outage authorisation processes.
9. **Transmission Line Configuration**
   Setup of transmission lines, line relationships, and ownership zones.
10. **Conflicting Line Configuration**
    Configuration of conflicting line rules used for outage validation.

4.0 LIST OF USERS ROLES

No	Name	Division
1	Requestor	
2	Planner	
3	GNM ADMIN	
4	GNM	
5	GNC	
6	Sysadmin/Master Admin	
7	Project Engineer	

5.0 USER REQUIREMENT – System setup

5.1 System Setup – Master Admin

The Master Admin or System Admin will have the responsibility of setting up new users, roles and permissions in the ICOMS 2.0 system. The SysAdmin will have the freedom to create new User Roles and edit the permissions of the existing roles within the system.

For each type of User Role created, there will be a set of permissions. Permissions will be configurable via the Admin UI and stored in the database; default role templates will be seeded at installation.

The system will be seeded with the Roles and permissions with this document upon the creation of the system.

During the creation of the Users, they will be assigned a Zone. The system features role based and location-based security. All users of the same role and location will be able to view and edit each other's data. They will not have permission to edit data of zones they are not assigned to.

The organisation dropdown will show the unique organisations created for external and external users based on the role selected. All external user will be given the role GCU (Grid Connected User). If GCU is selected as role, there will be a new field for users to select the GCU Type.

When a user is to be transferred to a different role or location, they will need to reach out to the SysAdmin to made the required changes to their role and location. This can be done within the system with a Role update feature. This will require the user to enter the new location they want access to along with a request summary which needs to be filled out with the details of the transfer.

Field	Field Type	Remarks
TNB ID	Integer	TNB ID for users in the Active Directory for internal users
Full Name	Text Field	User full name / external user PIC full name
Email Address	Text Field	Email address for internal and external users
Phone Number	Text Field	Phone number for all users
Role	Dropdown	Taken from list of all roles created
Assign Zone	Dropdown	Taken from list of all zones created
Organisation	Dropdown	List of all organisations in the system. Unique for internal and external users. List is also based on zone selected.
GCU Type	Dropdown	Shows only if role GCU is selected. Taken from GCU type dropdown list.
GCU Stations	Dropdown	Shows only if role GCU is selected. Shows list of all stations in the system, which will be assigned to the GCU user for outage creation

Table 5.1a Field for New User Creation

The system will be linked to the TNB active directory, allowing users access to the ICOMS 2.0 system with the same username and password as stored in the directory. ICOMS system will use the TNB ID number to link with the active directory.

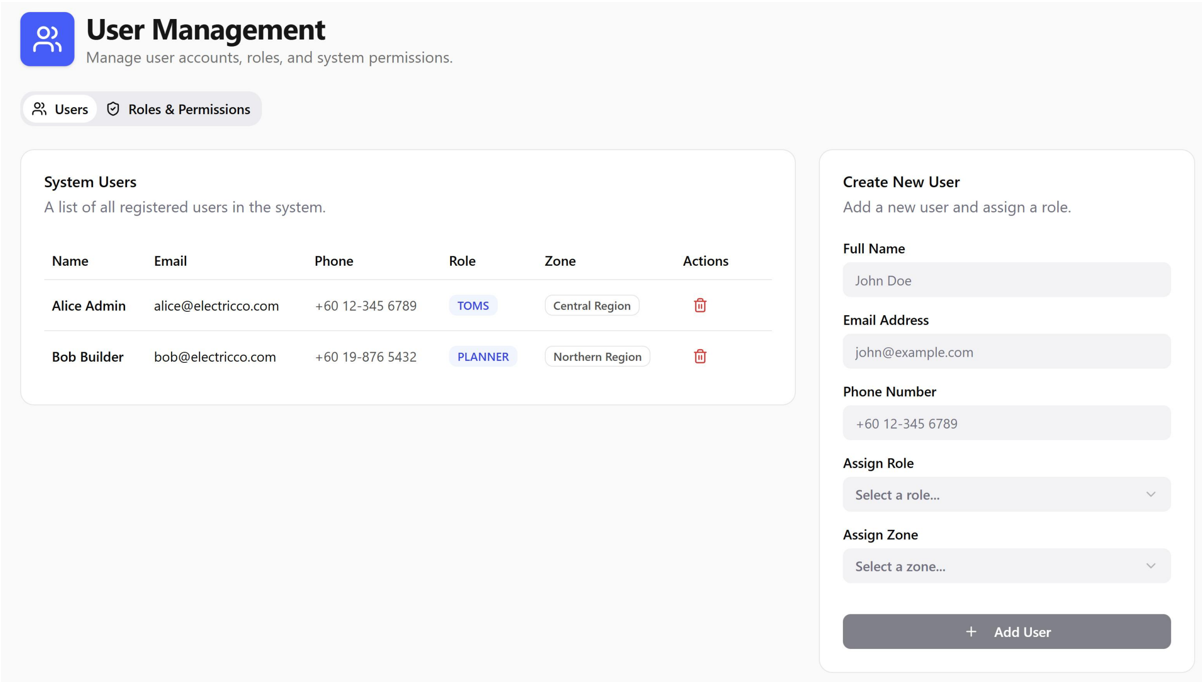

Figure 5.1a User Creation

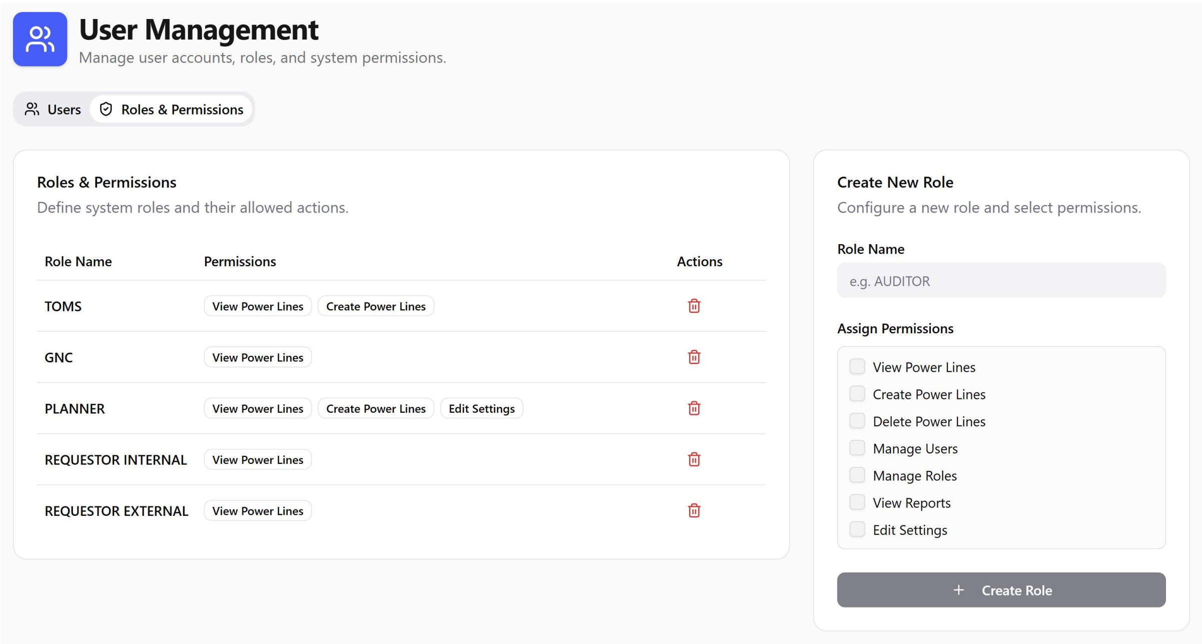

Figure 5.1b User Role and Permission setup

5.2 System Setup – System configuration

The GNM ADMIN along with input from other specific admin users will be responsible for the configuration of the system data. The entire system will pull data from these listing which will be maintained and updates by the designated Admin users.

5.2.1 Zones and Location – GNM ADMIN

The system will feature a role-based security, which allows users of the same role to view and edit data entered within the same zone.

GNM ADMIN can create Zones and map states/federal territories to Zones. The system will include the following default locations: Johor; Kedah; Kelantan; Melaka; Negeri Sembilan; Pahang; Perak; Perlis; Pulau Pinang; Selangor; Terengganu; Kuala Lumpur. Admins may edit mappings.

The current Zone setup is as follows

Zone	Locations
KL	Kuala Lumpur
Selangor + Negeri Sembilan	Selangor, Negeri Sembilan
Eastern	Terengganu, Kelantan, Pahang
Northern	Perlis, Pulau Pinang, Kedah
Perak + Melaka	Perak, Melaka
Johor Bahru	Johor

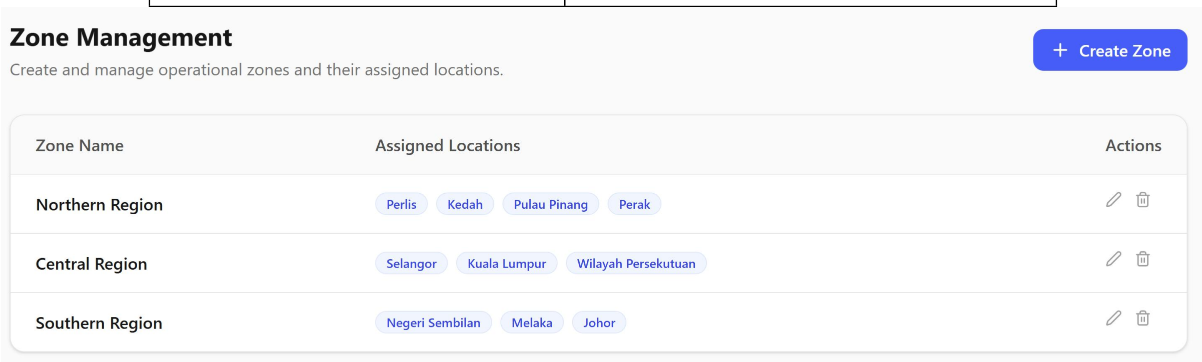

Figure 5.2.1a List of Created Zones

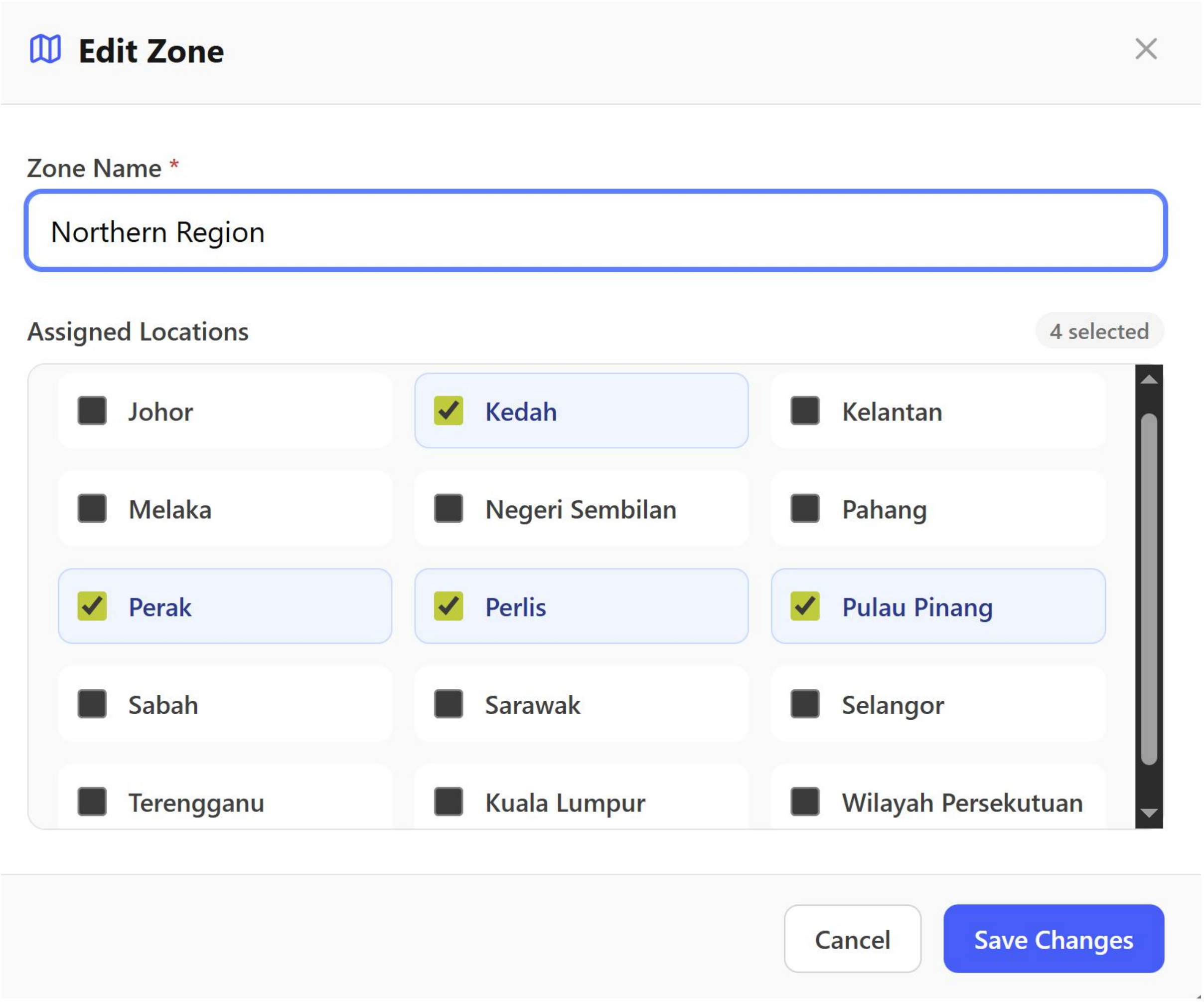

Figure 5.2.1b Creating and Editing Zones

5.2.2 Organization and Station Management – GNM ADMIN

Within each Zone created by the user, they will now be able to create different Organizations and Stations locked to those zones. Whenever the created Zone is selected throughout the system, it will populate the Organizations and Station fields with the data assigned here.

For all stations and originations, there will be an Abbreviation. This short-hand will be widely used within the system and needs to be unique. No two Station Name, Organization Name and Abbreviations can be the same.

There will be two types of organisations. Internal and External. This unique fields will be shown only during user creation, to distinguish between the two user types. When creating each organisation, the user will need to use a checkbox to select if the organisation is external or internal.

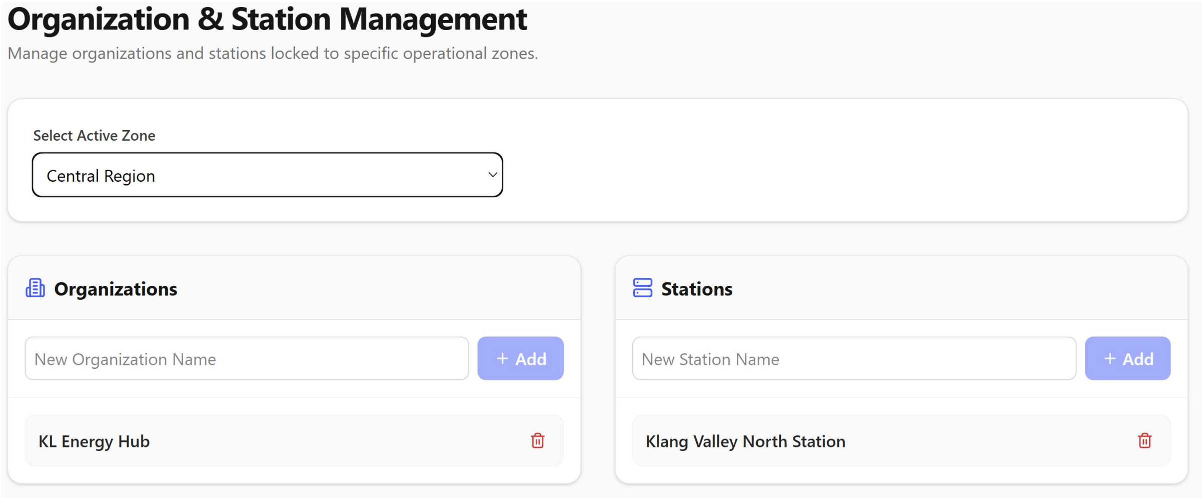

Figure 5.2.2a Creating originations and Stations per zone

5.2.3 Voltage and Equipment Management – GNM ADMIN

The GNM ADMIN user will be required to create different Voltage levels. These Voltage levels will be parked under all Stations across all Zones. The user will not be required to create Voltage levels individually for each Station.

For Each Voltage Level, there will be different Equipment Types. Users will only be able to see the Equipment type of the selected Voltage whenever prompted. Equipment type will be a system wide creation as well, linked only to the Voltage level. Equipment created of a specific equipment type and voltage will be linked to the stations.

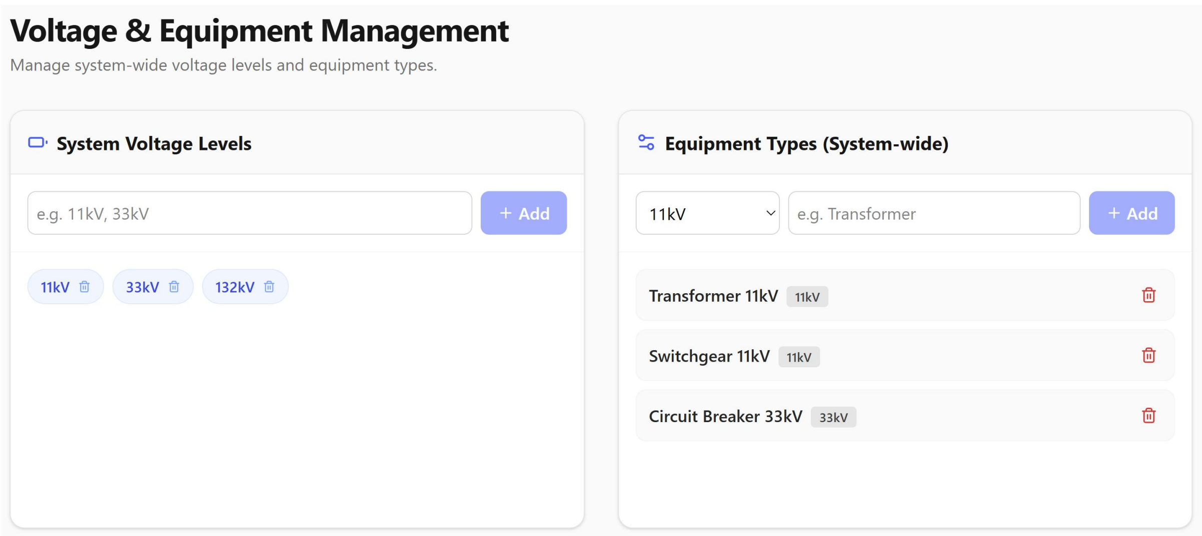

Figure 5.2.3a Creating System wide Voltage levels and the Equipment type for each Voltage

5.2.4 Equipment Creation – GNM ADMIN

Once the Voltage and Equipment Types have been set up, the user is now able to create individual equipment within the system, each one tied to a certain Type, Voltage, Station, and Zone.

Each equipment created will have a History section, where every outage the equipment has undergone will be stored. Users will be able to view the Outage number, Job Type, Date and time along with all Under- Study notes (Under- Study, Remarks, Highlights) filed. This information will be displayed in other sections of the system.

When creating Equipment, there will be two additional options. Boolean choice for Equipment position (Open/Closed – by default it is set to Closed) and a check-box for Off-Point Status. If the Off-point Status is checked, the Equipment Status automatically sets to OPEN, and this Equipment is set into a separate Off-Points module listing screen. While the status for all Equipment can be set to Open or Closed, they will be seen as temporary Off-Points being shown in a different tab on the Off-Points listing screen. Only the equipment with the Off-Point check-box marked as treated as permanent Off-points.

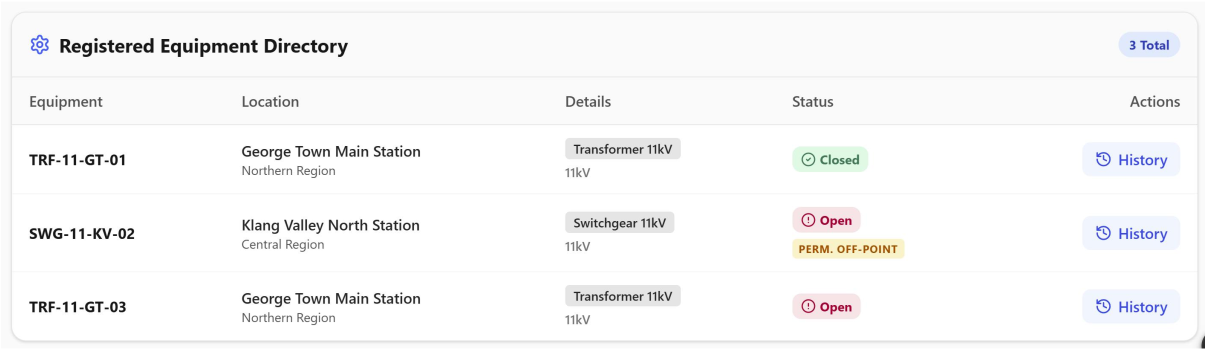

Figure 5.2.4a Equipment Listing with History Access

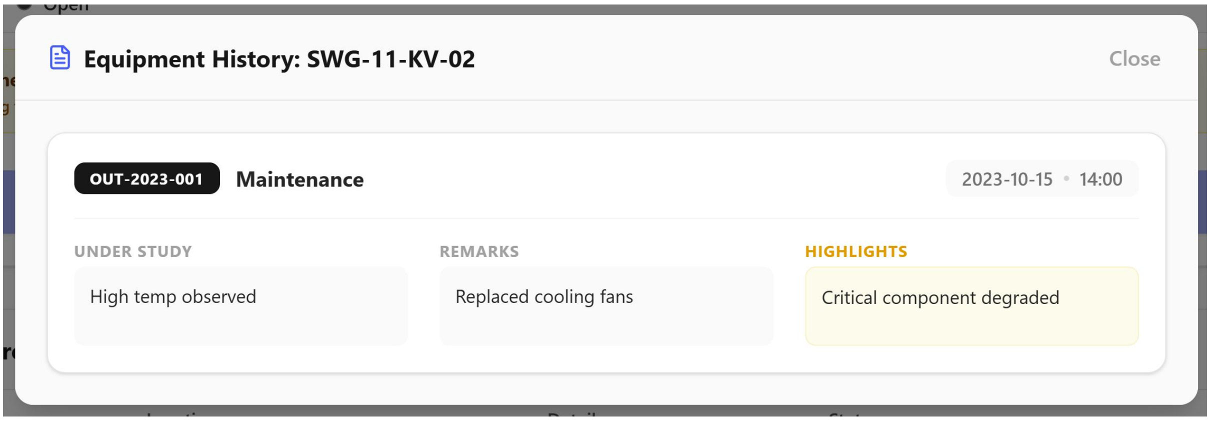

Figure 5.2.4b Individual history for each equipment

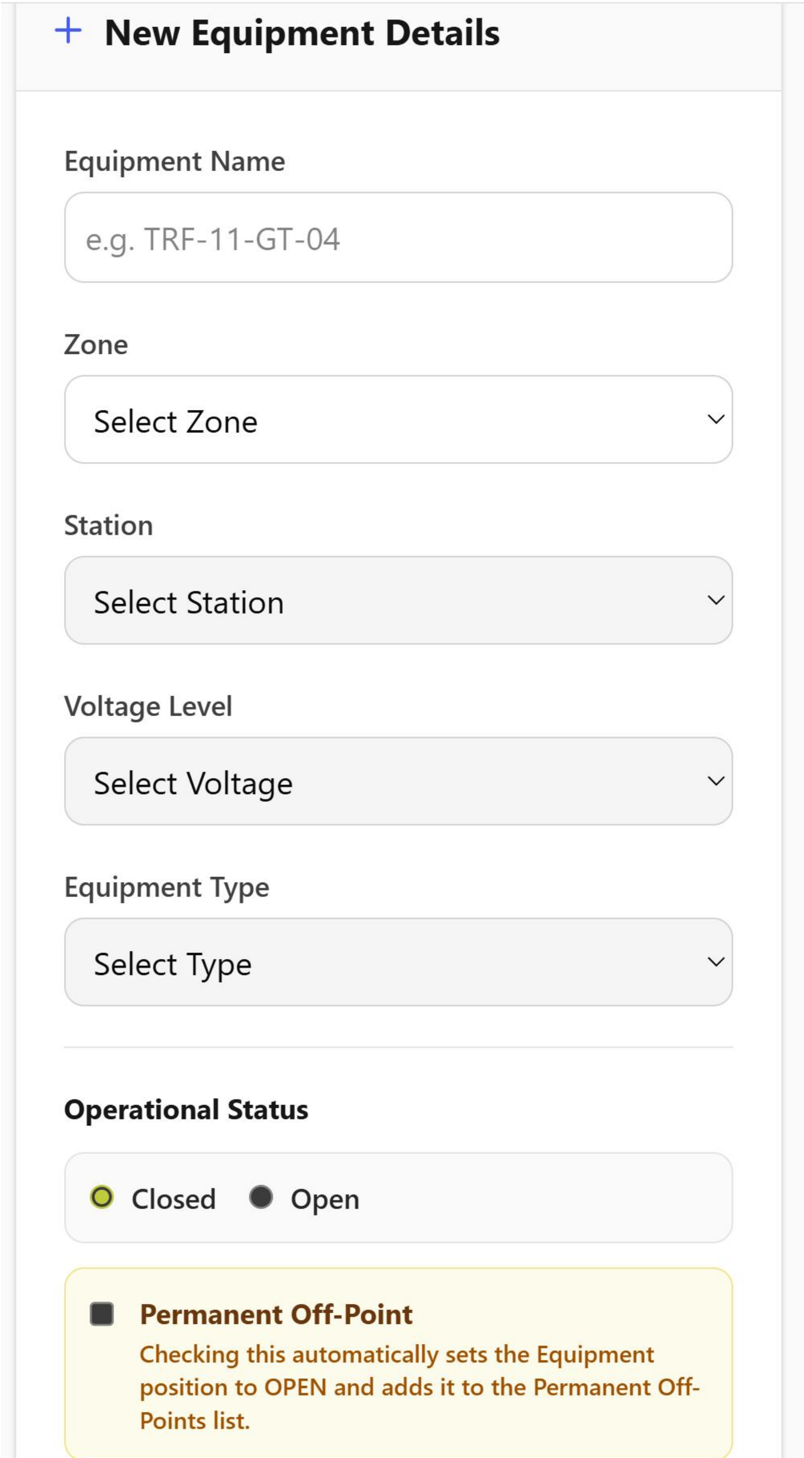

Figure 5.2.4c Creating New Equipment

Zone A	Zone B	Zone C
Org A,B,C	Org D,E,F	Org G,H,I
Station 1,2,3	Station 4,5,6	Station 7,8,9
Voltage Level		
Equipment Type per Voltage		
E1,E2,E3	E4,E5,E6	E7,E8,E9

Table 5.2.4 An example of Zones and equipment relationship

Name convention is done automatically. It will follow the following formulae

Voltage Level – MVA – Text (equipment name)

5.2.5 Off-Point Management – GNC Admin

There will be 2 tabs on the listing screen showing Permanent and Temporary. - All equipment marked as permanent Off-Points during equipment creation appear in the Permanent tab; equipment temporarily set to Open during an outage appear in the Temporary tab. Once the Equipment position has been normalized back to Closed, it will be removed from the Temporary tab.

GNC Admin users will be able to manually create Off-Points out the Equipment Creation screen. The create New Off-Point will allow users to select the Zone, Station, Equipment Type, and the Equipment Number; then mark it as an Off-Point.

During creating, users will be able to add remarks for all Off-Points, permanent and temporary. These will be shown in the Off Point management. Users will be able to filter the off points by region.

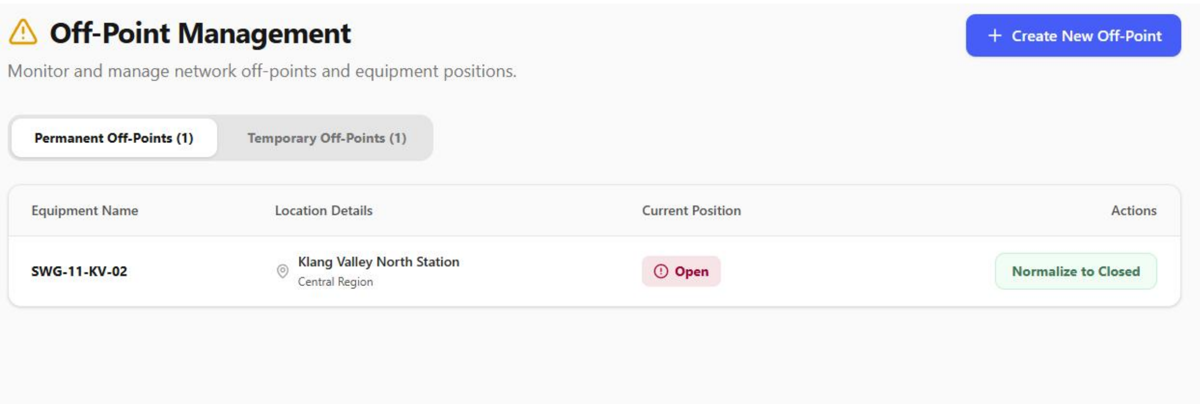

Figure 5.2.5a Off Point listing, Permanent and Temporary

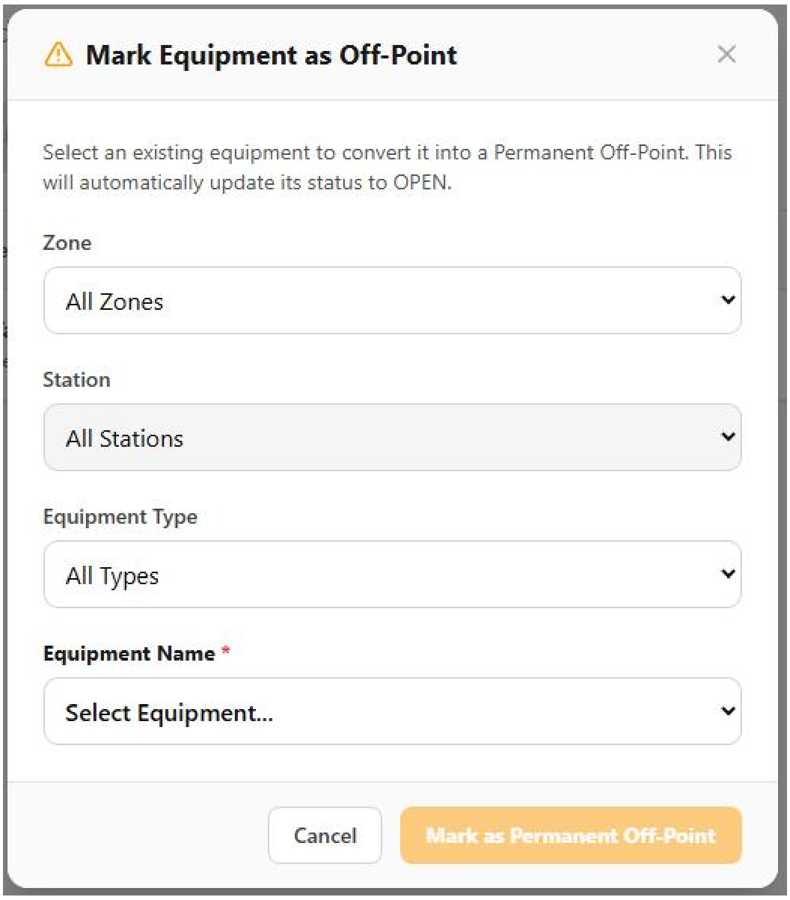

Figure 5.2.5b Setting new Equipment as Off-Points

5.2.6 Dropdown Management – GNM ADMIN

There are multiple selections for different field during an outage creation which will be managed by the Admin user. This will be a list of options for the dropdown which the Admin will be able to amend as required. Each dropdown option will be prefilled by the current systems information.

Admin should also be able to control the position of each option in the dropdown list.

Job Types will be sorted by the different types of Outages (Planned, Unplanned, Emergency, Forced). This will allow certain type of outages, such as Tripping to only be shown in the correct type of outage.

Dropdown	**Options**
Job Type (by outage type)	Conditional Monitoring; Defect Correction; Fault Investigation; Inspection; Routine Maintenance; SCADA & SAS; Testing; Projects; Others; Distribution Work; Transfer of Asset; Repair and Rehabilitation
Work Type	Live; Dead
Sequence	One at a time; All at the same time; Not applicable
Restoration	Immediately; 15 Minutes; 30 Minutes; 45 Minutes; 1 Hour; 1.5 Hours; 2 Hours; More than 2 Hours; Not Applicable

Not-Taken Reasoning	*List of Reasons why the Outage was not taken to be selected by outage Requestor*
GCU Type	Data Centre; Large Scale Solar; Large Power Consumer
MVA Rating	240 MVA, 180 MVA, 90 MVA, 245 MVA, 30 MVA

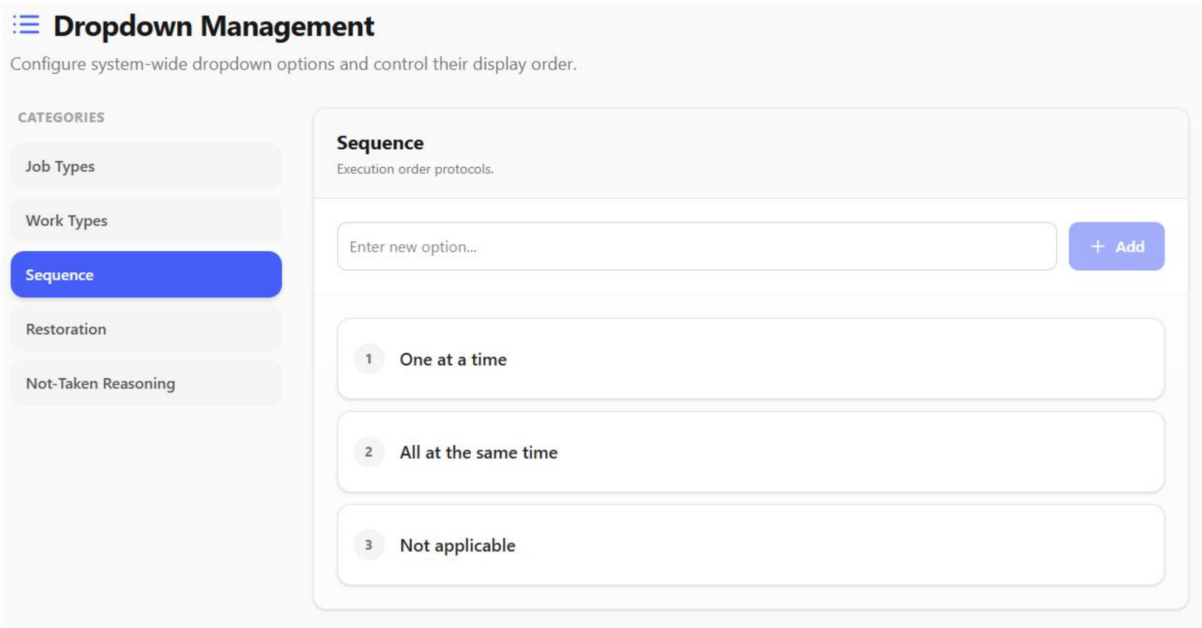

Figure 5.2.6 Dropdown Management

5.2.7 Project Management – Requestor/Planner Admin + GNM ADMIN

If any Job Type expect Maintenance is selected, the users should be able to select the project they are tying this outage to. This is set up by a Project Engineer role, assigned to a Requestor or Planner. The Project Engineer is responsible for the creation of projects. There only field needed is the Project Name. Once the Project has been created, it will over time be populated by outages for the Project. The GNM ADMIN has the responsibility of marking a project Complete. Once a Project is marked Complete, it is no longer available in the Project dropdown selection.

Field	Field Type	Remarks
TP Code	Text Field	Unique TP ID
Name	Text Field	A project name that can be reused
Project Name	Auto-filled	TP Code - Name
Start Date	Auto-filled	The start date of the first outage created for this project
End Date	Auto-filled	When TOMS closes a project
Status	Boolean	Open/Closed

If the GNM ADMIN tries to mark a Project with the Completed status while there are still open outages within the Project, the system will show a warning and require conformation before status change. If the status change is confirmed, the open outages will proceed as normal, but no new outages may be marked to that project.

TOMS may re-open a project by unchecking the Completed status box, which will again show a warning and will require conformation of status change.

Project names are created with a unique prefix TP Code followed by a reusable suffix

E.g. “LEVEL 4 TP CODE – PROJECT NAME”
In the above example, LEVEL 4 TP CODE is a unique entry that cannot be used again, the POREJCT NAME can be reused multiple times.

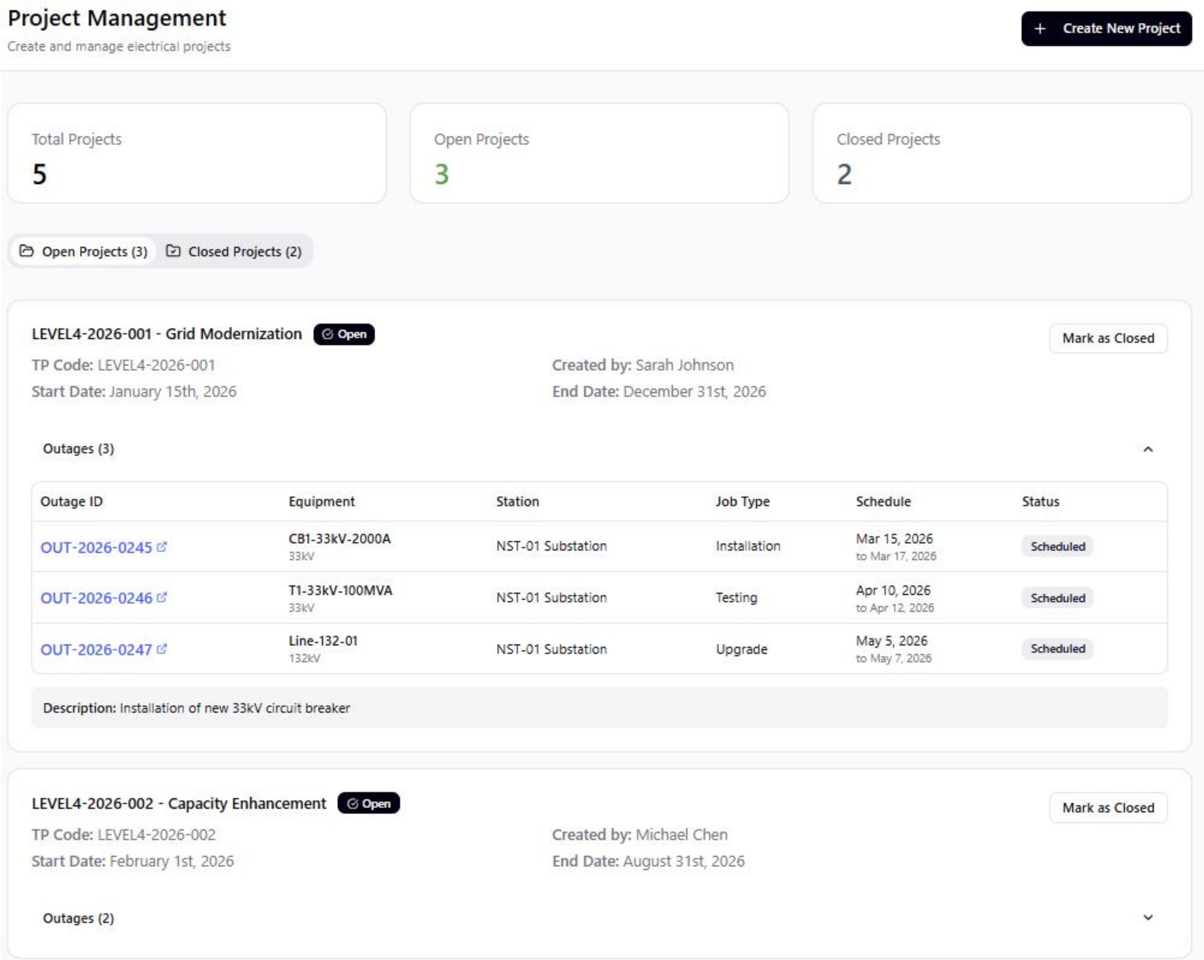

Figure 5.2.7a Project Management Screen

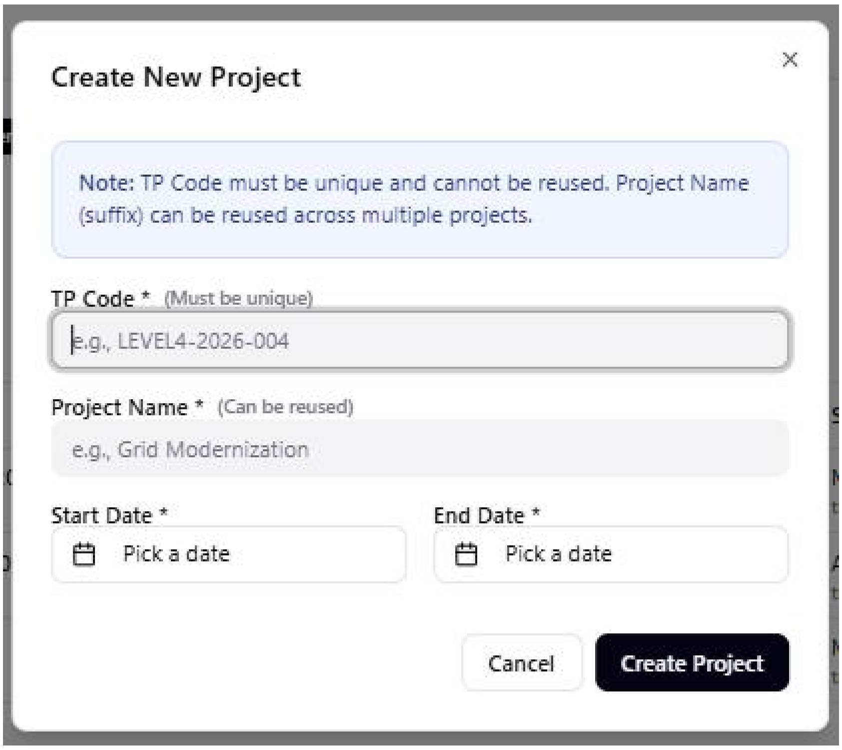

Figure 5.2.7b Create new Project

5.2.8 Outage Type Configuration – GNM ADMIN

There are 4 outage types within the system.

- Planned – Created by Requestor/Planner/TOMS. Requires Agreed (Planner) and Confirmed (Requestor) statuses before TOMS Approval.
- Unplanned - Created by Requestor/Planner/TOMS. Requires Agreed (Planner) and Confirmed (Requestor) statuses before TOMS Approval.
- Emergency - Created by Requestor/Planner/TOMS. Does not require Agreed (Planner) status. Requires Conformation (Requestor) status before Approval.
- Forced – Created by GNC.

All outages created by TOMS are marked Agreed and Confirmed automatically, and pushed to pending stage.

With the enhancements of ICOMS 2.0, users will not be able to select the type of outage being created. The outage type will be automatically selected based on the timeline of the outage during creation.

The outage time configuration will be set up by GNM ADMIN. If outage duration falls within the specified range for the selected voltage and work type, the system auto- assigns the outage type.

	Work Type: Dead Outage						
	More than (Days / Months / Years)			Less than (Days / Months / Years)			Voltage Multi-Select
	Day	Months	Year	Day	Month	Year	
Planned		6				3	All Voltages
Unplanned	7					1	132kv

Unplanned		1				1	275kv
Emergency	1				1		275kv, 500kv
Emergency	1			7			132kv
Forced				1			All Voltages

	Work Type : Live Outage						
	More Than			Less Than			Voltage multi-select
	Day	Months	Year	Day	Month	Year	
Planned		1					All Voltages
Unplanned					1		All Voltages

For each Work type (Dead and Live), users will be able to create multiple outage rules depending on the Voltage selected. When the outage time is set during the outage creation, it will auto populate the Outage Type based on these rules. The system will detect if rules are clashing with each other, and will show a warning. Users will have to create unique rules for the system to function.

I.e. if the Planned for All Voltages is more than 6 months and less than 3 years, a conflicting Unplanned rule for more than 6 months and less than 3 years cannot be set up.

5.2.9 Outage Scheduling – GNM ADMIN

In the current ICOMS workflow, Planned outages are only available from January to June. To allow users to control this feature, ICOMS 2.0 will allow GNM ADMIN to set up for each outage type the months the outage can be created.

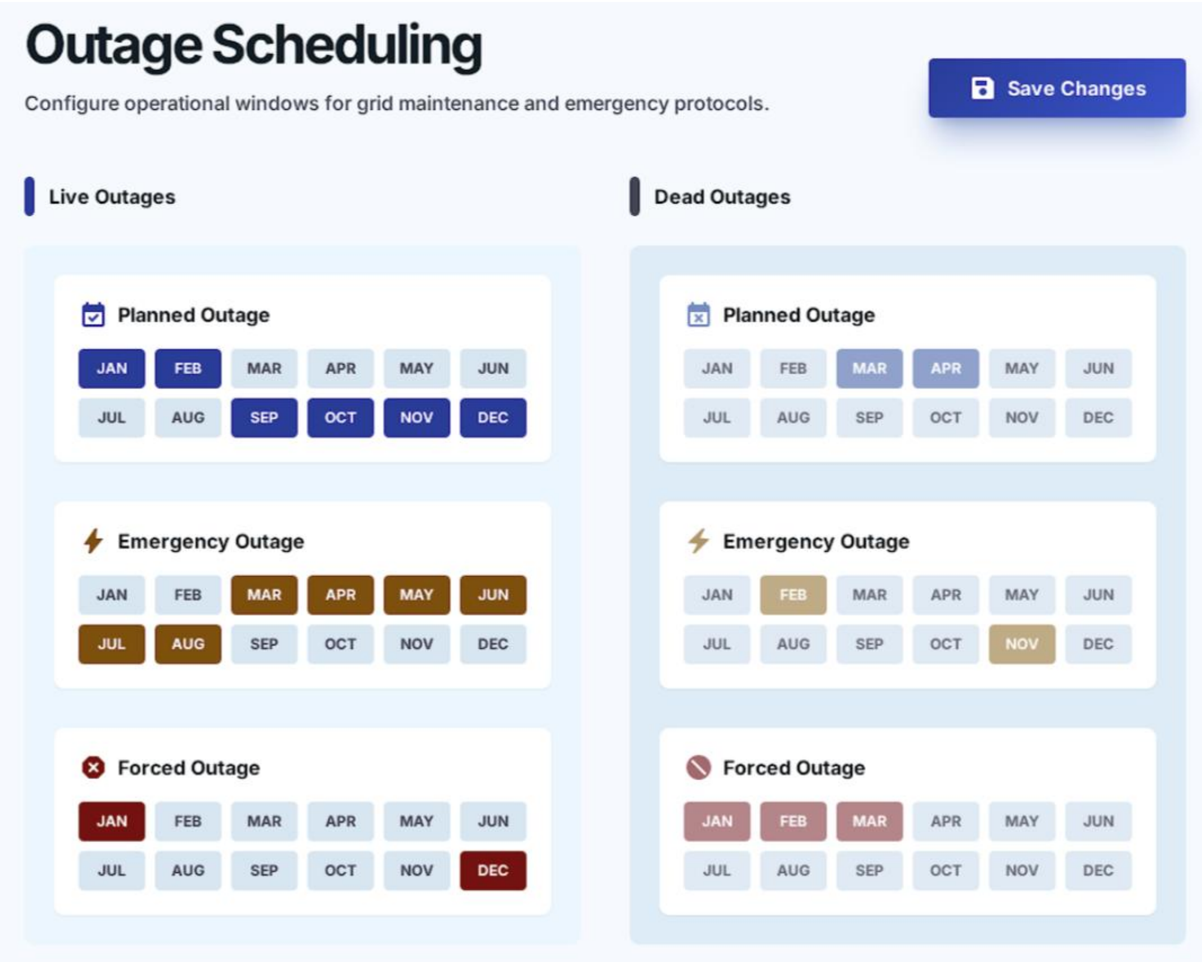

Figure 5.2.9a Outage Scheduling setup

5.2.10 Authorisation Personnel List – GNM ADMIN

While the outage is being fulfilled by the GNC team, they will need to Authorize and Unauthorize the outage, which will give the status of Taken-Active and Taken-Completed respectively. This is done by selecting an Authorisation user, along with the current time. Once Authorisation, the system will send an email to the Authorisation user notifying them of the outage Authorisation. TOMS will maintain this list of Authorisation Personnel, which will be based on User Roles that are assigned to the senior members of the GNC team.

Email Template – Email to user upon GNC Authorisation and Un-Authorisation

Subject	Outage {outageID} Authorisation
Email	Outage {outageID} was Authorised by your credentials at {autTime} by {userID}. This is an automated email by ICOMS 2.0.

Subject	Outage {outageID} Authorisation Removed
Email	Outage {outageID} has been Un-Authorised by your credentials at {autTime} by {userID}. This is an automated email by ICOMS 2.0.

5.2.11 Setup Lines – GNM ADMIN

ICOMS 2.0 will feature a new setup methodology for creating lines within the system. User will select the number of stations in the Line setup, which will automatically set the line type as either Single Line or Line Tee-Off

Once the number of stations is selected, the Station details will need to be filled out by selecting the Station Abbreviation, along with the Naming integer and the Line Number.

Once these details are filled out, the system will auto create the Line naming conventions, which when agreed by the GNM ADMIN can be saved within the system.

The naming convention is as follows

Single Line

Naming Integer- {Abbreviation of station 1}- {Abbreviation of Station 2}-Line number

These formulae will auto populate the line naming convention and save it. The line name will invert for the other station. The station 2 will come first and then station 1. If station 1 is selected in outage request, the line will be shown from station 1. If station 2 is selected in outage request, the line will be shown from station 2.

E.g.

123SDNK-GPTH1 (Shown if SDNK station selected)
123GPTH-SDNK1 (Shown id GPTH station selected)

Line Tee-Off

Naming Integer-{Abbreviation of stations 1}-{Abbreviation of Stations 2}-Line number/{Abbreviation of Stations 3}

In case of three or more stations, the first and last station will be taken as a Single Line, with the others at a Tee-Off position between them. The naming convention for the Tee-Off station will be done alphabetically. The system should allow up to 5 lines to be created, following the same naming convention.

E.g.

123SNDK-GPTH/SIDC1 Station 1 – Station 3/Station 2
123GPTH-SNDK/SIDC1 Station 3 – Station 1/Station 2
123SICD-GPTH/SNDK1 Station 2 – Sation first alphabetically/Remaining Station

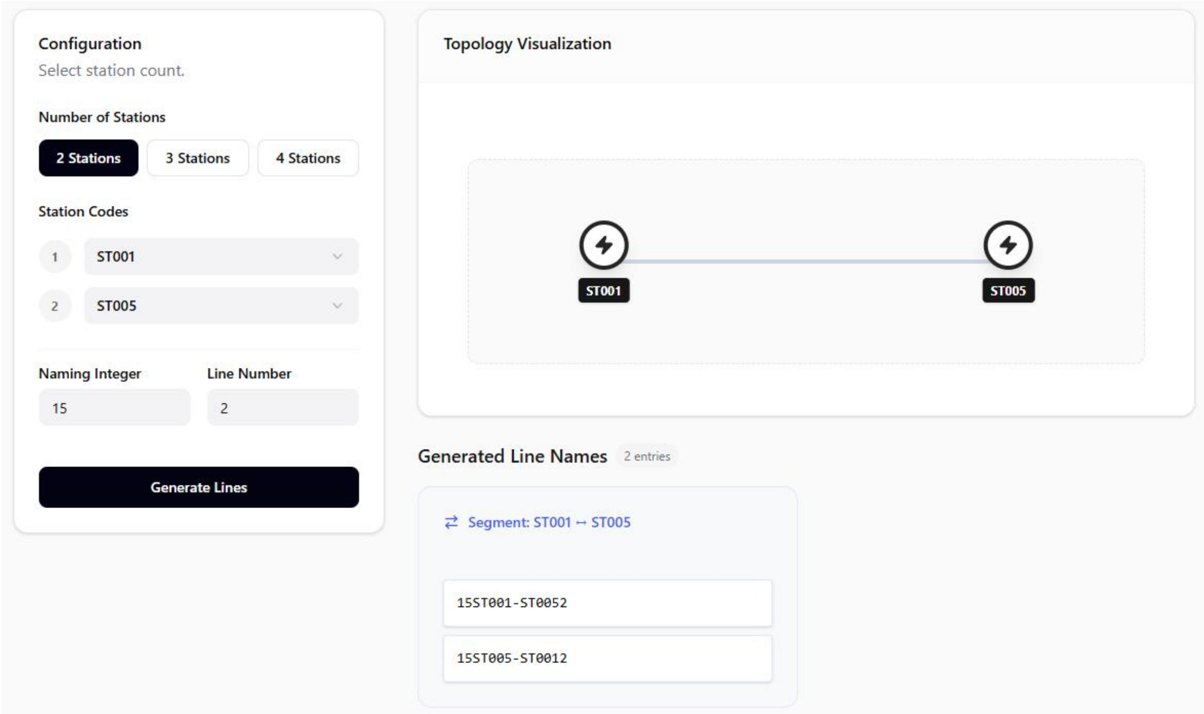

Figure 5.2.11a Setup of Lines and Station

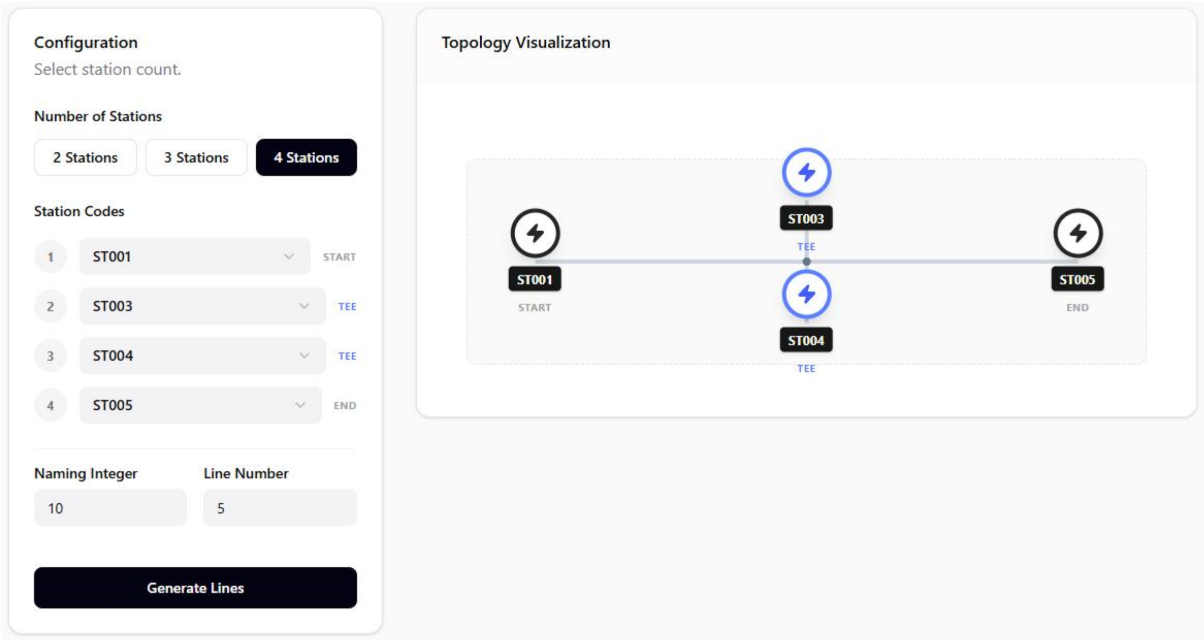

Figure 5.2.11b Visual Representation of 4 stations in Tee-Off

Each line will have an "Owner" region. By default, this is the Station 1 Zone. GNM ADMIN can add other "Owner" Zone to this line. This will allow users across multiple Zone access to this line for outage requests. Only users registered on the Zone will be able to apply for outage on those lines.

5.2.12 Conflicting lines setup – GNM ADMIN

Once the lines have been setup by the GNM ADMIN, they can proceed to setup the Conflicting lines rules. This will be two Line Equipment IDs which when conflict with each other if the outages for the lines are planned on the same time. This data will be used to show warning during the outage creation.

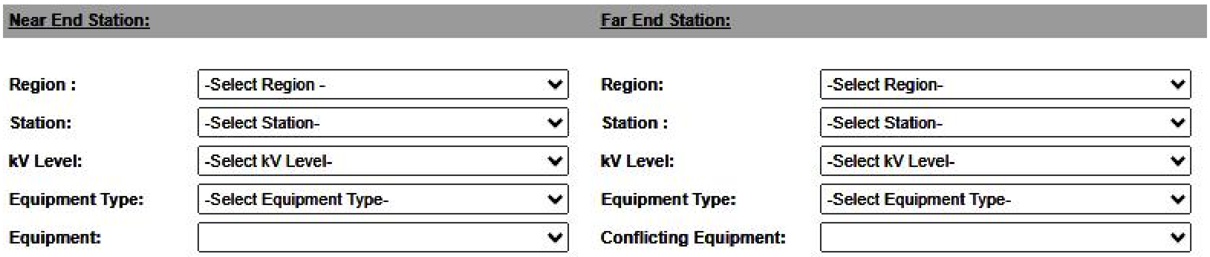

Figure 5.2.12a Setting up Conflicting Lines

5.2.13 Linking lines setup – GNM ADMIN

As with conflicting lines, users can now set up Linking Lines. There are the lines in a quad tower setup. During outage creation, if a line is selected that has a linked line, the linked line will be shown in the additional equipment selection. This will cause both linked lines to be scheduled for outage at the same time.

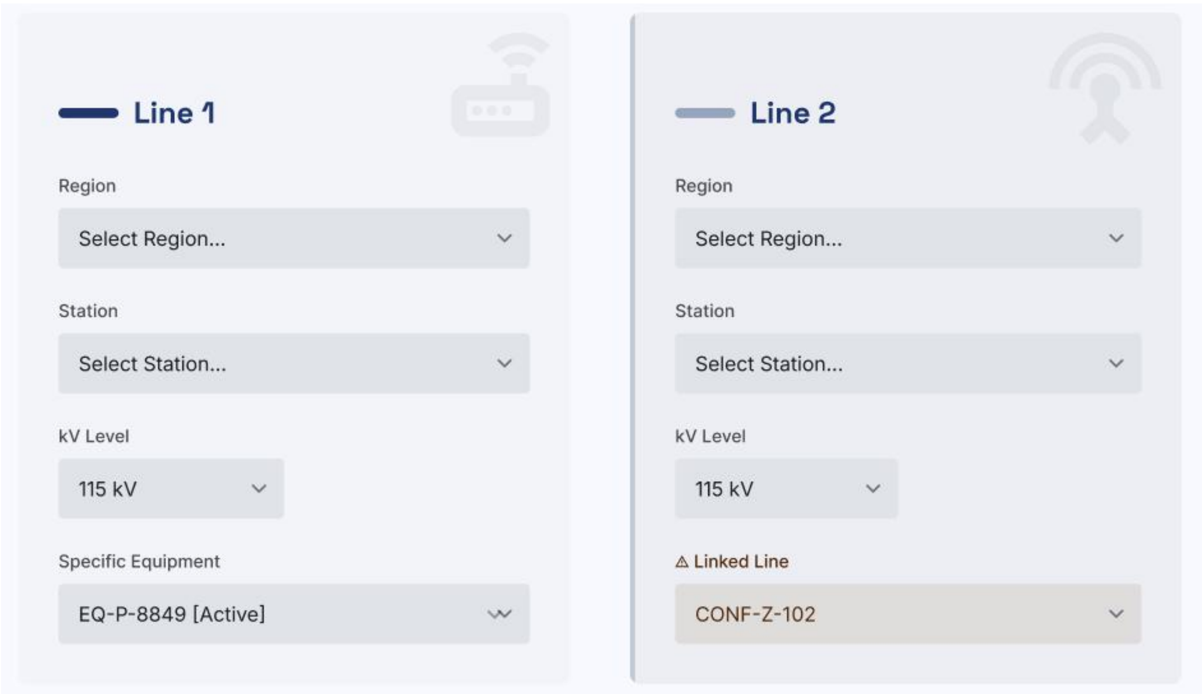

Figure 5.2.13a Setting up Linking Lines

5.2.14 Change request timeline setup – GNM ADMIN

If an outage has not been approved by the GNM team and the outage date/time has started. REQUESTORs will be allowed to submit a change request for a set number of days before the outage is pushed to the KIV list. The GNM ADMIN will be responsible for setting up the number of days a change request is allowed before the outage is pushed to KIV.

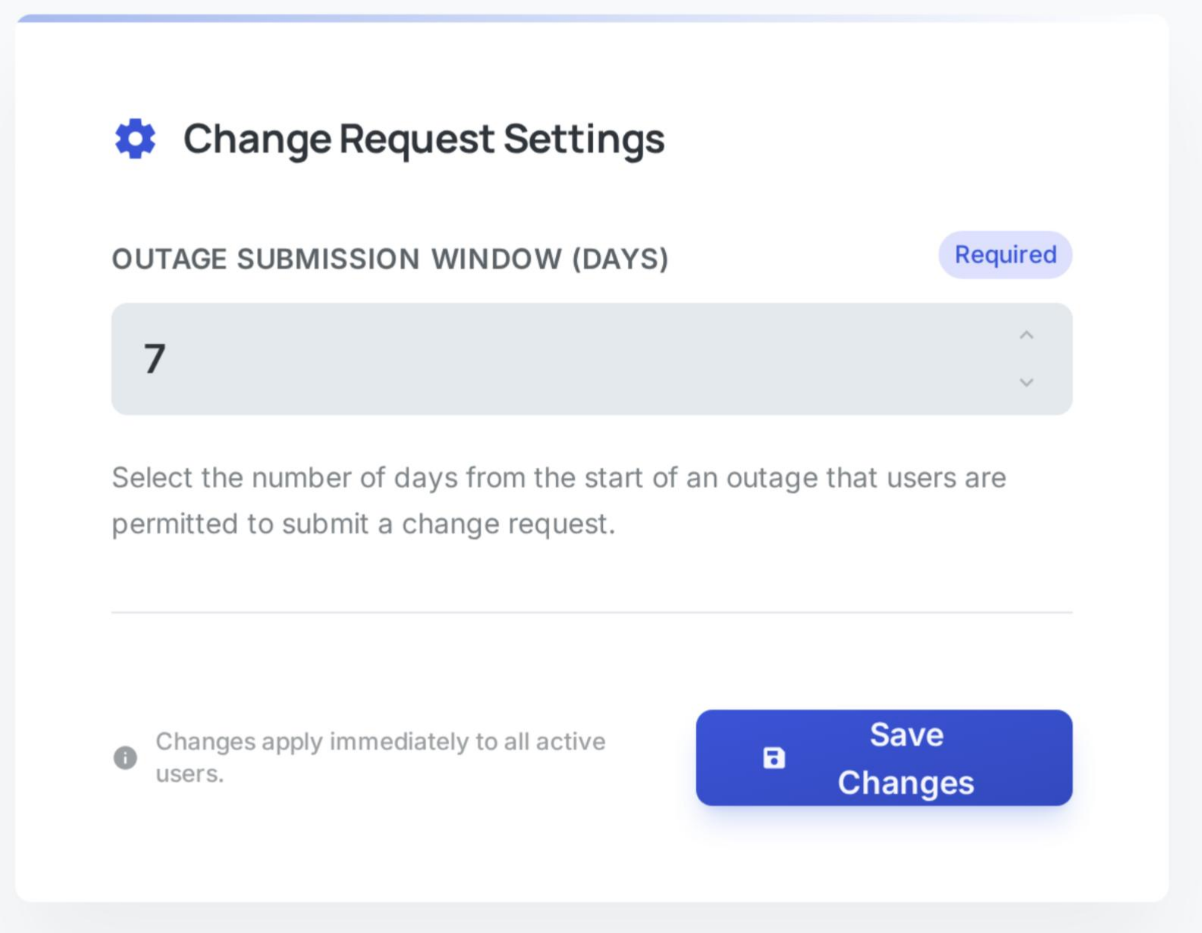

Figure 5.2.14a Setting up Change Request timeline

5.2.15 Add Mnemonic List – GNM ADMIN

The GNM Admin will be responsible for keeping the Mnemonic list up to date by uploading the latest copy upon revision. This document will be accessible by the GNM Engineer during SLD setup.

5.3 User Account

Each user in the system will have a User Account where they will have limited access to the Account information with the ability to update.

Name	Abdul Rahman	**Edit**
Role	TOMS-ADMIN	**Request Change**
Zone	KL	**Request Change**
Email Address	abdulrahman@tnb.com.mu	**Edit**
Phone Number	123456789	**Edit**
Password	*********	**Change Password**

Users will not be able to change their Role or Zone; however they can request a change which will send an email to the SysAdmin with the details and reasoning by the user.

Subject	{userID} requests change in User Role/Zone
Email	{userID} has requested a change in their User Role/Zone to {newRole} + {newZone} with the following justification. {changeComment} This is an automated email by ICOMS 2.0.

Users can edit their name, email address, and phone number at will.

For external user, to change password, the user will need to input the Old Password along with two iterations of the new password.

If the user has forgotten their password, they will reset the password from the login screen, which will send an email to the registered email address with the link to setup a new password.

Internal users will need to refer to the Active Directory for password change and recovery.

6.0 USER SIGN OFF – System setup

User #1 Signature: Name: Designation: Dept.: Date:	User #2 Signature: Name: Designation: Dept.: Date:
User #3 Signature: Name: Designation: Dept.: Date:	User #4 Signature: Name: Designation: Dept.: Date:
User #5 Signature: Name: Designation: Dept.: Date:	User #6 Signature: Name: Designation: Dept.: Date: